Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space. Because sensor telemetry contains unforecastable jitter and scale disparities, observation-space residuals develop heavy tails that trigger severe false-alarm flooding under automated calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid reconstructing raw noise by forecasting compact latent summaries of future windows, yet collapsing high-dimensional multi-horizon prediction errors into a single scalar anomaly score introduces a second bottleneck: one or two out-of-sample drifting coordinates can dominate the scalar calibration tail, inflating the alarm threshold and suppressing point-wise F1. We introduce Soliton-Reynolds JEPA paired with a 224-bit multi-perspective fluid-dynamic circuit-breaker panel that eliminates scalar score collapse and parametric tail extrapolation. The architecture taps intermediate temporal convolutional stages, subtracts online context barycenters, whitens prediction errors across four nested sub-horizons, and expands each of the 32 latent coordinates into seven fluid-dynamic perspectives: four sub-horizon flow positions, Reynolds mean-flow transport, turbulent fluctuation amplitude, and convective right-edge acceleration, governed by calibration-stabilized whitening, per-dial hijack guards, and 32 fast right-edge onset relays. Across 33 primary and 141 archive-wide multivariate telemetry streams, latent prediction suppresses false alarms relative to observation-space reconstructors while the 224-bit circuit-breaker panel doubles strictly causal point-wise F1, lifts event recall to 72.7 percent, and cuts detection delay in half under both causal streaming and retrospective buffered attribution.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable healthcare [1]–[3]. In NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels track power buses, thermal gradients, vibrations, and kinematics [4]. Monitors must identify subsystem faults rapidly without overwhelming operators with nuisance alerts.

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

Twin Bottlenecks: Observation-Space Noise Flooding and 1D Scalar Score Collapse: In deployed monitoring, an automated detector must trigger binary alarms using healthy (nominal) calibration data alone. Standard deep detectors (TimesNet, TranAD, Anomaly Transformer) reconstruct raw sensor readings and flag timestamps with large point-by-point error [5]–[7]. Because raw telemetry contains sensor jitter and scale differences across channels, point-level errors produce frequent, clustered spikes and a heavy-tailed residual distribution with high excess kurtosis ($\gamma_2 > 8$, Table I). Under automated calibration rules such as Streaming Peaks-Over-Threshold (SPOT) [8], normal background shifts easily cross the threshold, triggering false alarms on 22%–31% of healthy timesteps. Shifting prediction into a compact latent space suppresses sensor jitter, but existing latent detectors still collapse their multi-dimensional prediction residuals into a single 1D scalar anomaly score before calibration. In multi-horizon latent spaces ($4 \times 32 = 128$ whitened coordinates), if even one or two coordinates experience mild out-of-sample variance shift between training and calibration (*hijacker dials*), summing or gating all coordinates into a 1D scalar inflates the calibration tail and pushes the scalar threshold to the extreme tip of the distribution, causing the detector to fire only on isolated peak steps inside an anomaly.

Fluid-Dynamic Latent Prediction and the 224-Bit Circuit-Breaker Panel: Predicting a compact *summary* of a future time window rather than every noisy sensor sample filters out random jitter before the prediction error is computed. We study Joint-Embedding Predictive Architectures (JEPAs) [9], which compress the preceding 256 timesteps into a 32-dimensional summary vector and forecast the summary of the next S timesteps ($S = 64$ by default, or $S \in \{1, 4\}$ for fast tripwires) using healthy training data alone [10]. Alongside a single-horizon baseline (TS-JEPA), we introduce *Soliton-Reynolds JEPA*, which taps multiple depths of its temporal convolutional network [11], subtracts the running context barycenter to cancel slow level shifts, and whitens prediction errors across $M = 4$ nested sub-horizons. Drawing on Reynolds decomposition [12] and Lieb–Thirring spectral bounds [13], it separates multi-horizon errors into mean transport and zero-mean fluctuations. To eliminate 1D scalar score collapse and parametric tail extrapolation altogether, we pair the multi-horizon predictor with a *224-bit multi-perspective circuit-breaker panel* that expands the 32 latent coordinates into $7 \times 32 = 224$ fluid-dynamic dials (128 sub-horizon flow positions, 32 Reynolds slow-drift transport dials, 32 Reynolds

turbulent-fluctuation dials, and 32 convective right-edge acceleration dials) monitored by independent hijack-guarded binary switches and 32 fast right-edge onset relays. We evaluate four questions: (i) how much latent representations and the 224-bit circuit-breaker panel lower realized false-positive rates (FPRs) while improving point-wise F1, event recall, and detection delay; (ii) how normalization, learned latent targets, multi-horizon whitening, and multi-perspective Boolean tripwires each contribute; (iii) how detection varies across point spikes, contextual faults, and quiet single-sensor freezes; and (iv) how shortening S trades off alarm lag ($S-1$) against detection quality.

A. Contributions

1) *Zero-collapse 224-bit fluid-dynamic circuit-breaker panel:* We show that collapsing 128 whitened sub-horizon residuals into a 1D scalar score allows isolated drifting coordinates to inflate scalar thresholds and suppress causal Point-F1 (0.050–0.052). Replacing 1D score collapse with a parameter-free 224-bit circuit-breaker panel and 32 fast right-edge onset relays more than doubles causal Point-F1 ($0.052 \rightarrow 0.116$–0.138 on n=19 primary streams, $17/0$ wins, $p = 1.5\times10^{-5}$; 0.067–0.077 $\rightarrow$ 0.125–0.139 across n=141 archive streams, $114/18$ wins, $p = 1.2\times10^{-16}$), lifts causal event recall to 60.7%–72.7%, and cuts detection delay in half (Sections IV-D and VII, Tables II–V).
2) *Why latent predictors stabilize alarm decisions:* Across matched-backbone models (272k–312k online parameters), a layer-normalization (LayerNorm) [14] bottleneck compresses residual kurtosis from 6.10 to 0.91, predicting a learned latent summary cuts calibration-to-holdout drift in half (D_{KS} from 0.376 to 0.214), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to $D_{\mathrm{KS}} = 0.179$ while cutting the inter-block drift share of excess causal false alarms four-fold from 71.3% to 18.7% (Tables II(c) and V(b)).
3) *Fewer false alarms, contextual-anomaly dominance, and quiet-fault recovery:* Across the 19 non-GHL streams, latent prediction cuts held-out FPRs dramatically relative to TimesNet and TranAD (0.6%–1.7% causal and 7.8%–9.8% buffered vs. 22.2%–30.7% for reconstructors) while achieving 0.411 buffered precision-recall area under the curve (PR-AUC; 0.312 causal), leading on NASA contextual anomalies, and recovering quiet single-channel freezes when paired with a rolling sensor-variance monitor (Tables III and IV, Figs. 3 and 4).
4) *Multi-horizon wavepacket bounds, ablation controls, and fast response:* Proposition 1 proves that the Reynolds-centered Lieb–Thirring concentration ratio satisfies $\frac{1}{M-1} \leq R_{\mathrm{LT}} \leq 1$ with exact radial scale invariance, while fixed-backbone ablations show that multi-horizon whitened perspectives drive the primary gains (with post-hoc scalar wavepacket multipliers satisfying Two One-Sided Tests (TOST) [15] equivalence) and short horizons ($S \in \{1, 4\}$) eliminate the 63-step window lag (Table II, Fig. 4(a)).

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates the architectures and 224-bit circuit-breaker panel, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

TABLE I
KEY OPERATIONAL AND FLUID-DYNAMIC CIRCUIT-BREAKER TERMS.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 fluid-dynamic coordinate tripwires and 32 onset relays avoiding 1D score collapse.
Reynolds dials	Per-coordinate mean-flow transport $\lvert\bar{e}_c\rvert$ (32 dials) and turbulent fluctuation σ'_c (32 dials) across horizons.
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and convective acceleration/fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\mathrm{cal}}\gamma_j^{\mathrm{tr}\to\mathrm{cal}}$ preventing drifting coordinates from triggering false trips.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Excess kurtosis	Tail heaviness γ_2 of residuals; large values indicate frequent extreme spikes.
Soliton ratio	Bounded ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring whether sub-horizon fluctuations align into a rank-1 wavepacket.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [7] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [5] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [6] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger severe false-alarm flooding under baseline shift [16].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [17]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [18] and TS2Vec [19] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [20] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Table III(d)). Joint-Embedding Predictive Architectures (JEPAs) [9], [21], [22] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} produced by a separate

target encoder. Collapse is prevented via exponential moving average (EMA) target updates [23] or variance-covariance penalties such as VICReg [24], Barlow Twins [25], and batch whitening [26].

Fluid-Dynamic Decomposition and Multi-Perspective Circuit Breakers vs. Scalar Thresholding: Multi-sensor telemetry reflects physical dynamical trajectories [27]. Neural ordinary differential equations (ODEs) [28], Physics-Informed Neural Networks (PINNs) [29], and Hamiltonian or Lagrangian networks [30], [31] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [32]. Koopman and operator-theoretic models [33]–[37] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we draw on Reynolds decomposition [12] from turbulence modeling and Lieb–Thirring spectral bounds [13] to decompose multi-horizon latent residuals into mean transport, turbulent fluctuations, and convective acceleration. Instead, existing detectors collapse high-dimensional residuals into a 1D scalar score and extrapolate a threshold via Pickands–Balkema–de Haan extreme-value theory [38] and Generalized Pareto Distribution (GPD) fitting in SPOT [8]. On non-stationary telemetry with serial correlation [39] and coordinate-level regime drift, 1D score collapse allows a single drifting coordinate to hijack the scalar threshold. We replace 1D scalar collapse with a 224-bit multi-perspective Boolean circuit-breaker panel and evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [16], [40], [41].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [42], [43], TimeEval [16], and mTSBench [44], expanded to 141 qualifying multivariate streams across 12 datasets in our archive-wide evaluation. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains and separates natural operational or physiological anomalies from testbed or simulated faults. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [45] (K=38), all containing natural operational or solar-flare events; on SMAP and MSL, NASA's ground-truth archive [4] further classifies each anomaly sequence as either an out-of-range *point* fault or an in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [42] sampled at 64 Hz (K=9, 72 natural gait-freezing episodes, where S=64 equals a 1 s horizon). *Industrial processes and smart infrastructure* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 natural hydrological anomalies), Metro I-94 hourly highway traffic (K=5), Room-Occupancy sensing [46] (K=5), CalIt2 building flow [43] (K=2), and six CICIDS network streams (K=72, simulated attack flows). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [4], [16].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table III(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9, 1 held-out event) enters Cohorts 3–4. To eliminate subset-selection bias, Cohort 5 evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 Opportunity, GECCO, Genesis, Pooled Server Metrics (PSM), credit-card, Metro, and SWAN) at three seeds alongside the 130-stream baseline screen (Tables III(d) and V(a)).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [41]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Fluid-Dynamic Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and partitions the stream at index t into a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$). For $K > 1$, a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Channels are decomposed across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [11].

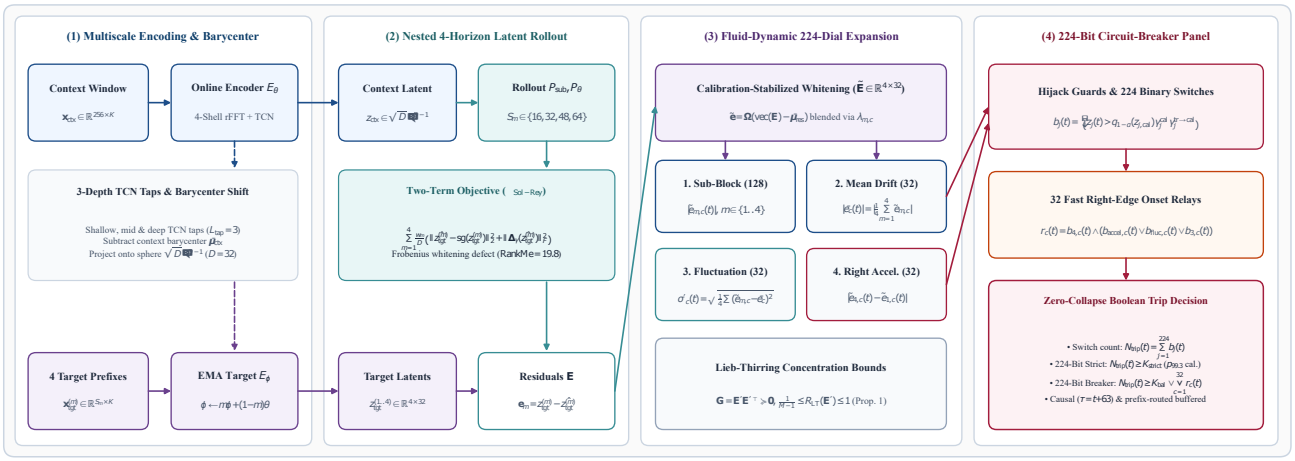


Fig. 1. Overview of the four-stage Soliton-Reynolds JEPA and 224-bit circuit-breaker pipeline: (1) multiscale dyadic windowing, 3-depth TCN hierarchy tapping, and barycenter centering; (2) nested 4-horizon latent rollout under the two-term whitening defect objective; (3) calibration-stabilized Mahalanobis whitening and Reynolds expansion into $7 \times 32 = 224$ fluid-dynamic dials; and (4) zero-collapse 224-bit Boolean circuit-breaker tripwires and 32 fast right-edge onset relays.

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit heavy-tailed sensor noise (Sections I–II), we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ onto the sphere $\sqrt{D}\mathbb{S}^{D-1}$, where target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with cosine momentum $m_k \in [0.995, 0.9995]$ [21], [23]. Because deep dilated TCN blocks smooth out sharp onset spikes while non-stationary streams shift the latent mean, Soliton-Reynolds JEPA introduces two structural operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* A tap head extracts $L_{\text{tap}} = 3$ intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic front-end, the mid-depth TCN block, and the terminal TCN block. For each tap ℓ, the channel-centered terminal state and temporal root-mean-square (RMS) energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, projected into $\mathbb{R}^D$, and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \le 0.5$:
$$\mathbf{z}_{\text{fused}} = (1-g)\,\mathbf{z}_{\text{deep}} + g\ \mathrm{LN}\big(W_2\,\phi_{\text{GELU}}(\mathrm{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$
2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updated at rate 0.05 on training batches is copied synchronously to the EMA target tap head and subtracted from both context and target fused vectors before terminal LayerNorm projection $z = \mathrm{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}}) \in \sqrt{D}\mathbb{S}^{D-1}$, canceling common-mode baseline drift between windows.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Covariance Whitening Defect Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with predictive alignment and VICReg variance-covariance penalties [24] ($\gamma_{\text{var}} = 1.0$). While terminal LayerNorm bounds representations on $\sqrt{D}\mathbb{S}^{D-1}$ and compresses residual kurtosis relative to observation-space reconstructors, pooling the entire $S = 64$ target window into a single vector averages out short-lived sub-horizon transients.

To resolve sub-window dynamics without sacrificing the 64-step lookahead, Stage 2 of *Soliton-Reynolds JEPA* partitions the target horizon $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ of lengths $S_m = (m/M)S \in \{16, 32, 48, 64\}$ (or $M = 3$ prefixes $\{16, 32, 64\}$ in our fast archive-wide variant *Speculative-JEPA*). The EMA target encoder E_ϕ encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \mathrm{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \dots, M-1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ forecasts $\hat{z}_{\text{tgt}}^{(M)} = \mathrm{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon mean-squared error (MSE) plus a Frobenius covariance whitening defect:
$$\mathcal{L}_{\text{Sol-Rey}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \mathrm{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big), \tag{2}$$
where $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ measures the whitening defect of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B\bar{\mathbf{z}}^\top$ with empirical covariance $\boldsymbol{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top\mathbf{Z}_c$ and marginal standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ ($d \neq j$). Unlike log-determinant barriers that become singular when batch size $B < D$, $\|\boldsymbol{\Delta}_\gamma(\mathbf{Z})\|_F^2$ remains smooth for any $B \ge 2$ and raises effective representation rank $\mathrm{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [47].

C. Stage 3: Calibration-Stabilized Whitening and 224-Dial Fluid-Dynamic Reynolds Expansion

Why does collapsing a multi-horizon residual tensor into a single 1D anomaly score degrade point-wise F1? Let $\mathbf{E}(t) = [\mathbf{e}_1(t), \dots, \mathbf{e}_M(t)]^\top \in \mathbb{R}^{M\times D}$ stack the $M = 4$ raw sub-horizon prediction residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ across

all $MD = 128$ coordinates. Fitting the Ledoit–Wolf shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [48] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows produces Mahalanobis-whitened residuals $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$ with unit training variance per coordinate. On unseen nominal calibration windows, however, one or two whitened directions often experience mild out-of-sample variance growth (*hijacker dials*) where $\mathbf{\Omega}$ amplifies small covariance shifts. When all 128 coordinates are summed or pooled into a 1D scalar score before calibration, those one or two hijacker dials dominate the upper calibration tail and push any scalar threshold to the extreme tip of the distribution, so that only isolated peak timestamps inside a true anomaly cross the threshold.

Stage 3 prevents both precision-matrix amplification and 1D coordinate collapse through two steps:

1) *Calibration-stabilized coordinate whitening:* On nominal calibration windows, we measure the whitened RMS $\text{RMS}_{\text{cal}}^{\text{white}}(m,c)$ and raw RMS $\text{RMS}_{\text{cal}}^{\text{raw}}(m,c)$ of each coordinate (m,c). Any coordinate whose whitened calibration RMS exceeds 1.10 is smoothly blended back toward its calibration-normalized raw residual $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\text{RMS}_{\text{cal}}^{\text{raw}}(m,c) + 10^{-4})$ via the rational stability weight $w_{m,c}$:
$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}_{\text{cal}}^{\text{white}}(m,c) - 1.10\big)^2}. \tag{3}$$
2) *Coordinate-wise Reynolds decomposition into $7D = 224$ fluid-dynamic dials:* In fluid turbulence, Reynolds decomposition [12] separates a flow field into its horizon-averaged mean transport and zero-mean turbulent fluctuations. Rather than collapsing across the $D = 32$ latent coordinates, we apply Reynolds decomposition along the lookahead axis $m \in \{1,\dots,M\}$ independently for each coordinate $c \in \{1,\dots,32\}$, expanding the stabilized residual field $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into $7D = 224$ non-negative physical-perspective dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$:
   (a) *Sub-horizon flow-position magnitudes ($4D = 128$ dials):* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ for $m \in \{1,2,3,4\}$, isolating fault magnitude at prefix horizons $\{16, 32, 48, 64\}$.
   (b) *Reynolds mean-flow transport / slow-drift dials ($D = 32$ dials):* $d_{\text{drift},c}(t) = |\bar{e}_c(t)| = \big|\frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)\big|$, capturing persistent trajectory displacement across all four prefixes.
   (c) *Reynolds turbulent-fluctuation / shear dials ($D = 32$ dials):* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M}(\tilde{e}_{m,c}(t) - \bar{e}_c(t))^2}$, capturing inter-horizon shear and oscillatory divergence.
   (d) *Convective right-edge acceleration dials ($D = 32$ dials):* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, measuring newly entering fault energy in the terminal sub-block $[48, 64]$ relative to the initial prefix $[0, 16]$.

Proposition 1 (Coordinate-Wise Reynolds Energy Orthogonality, Convective Shear Bound, and Lieb–Thirring Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top \mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M \bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise Reynolds energy orthogonality and acceleration bound: *For every coordinate $c \in \{1,\dots,D\}$, mean-square sub-horizon position energy decomposes orthogonally into mean-flow transport and turbulent fluctuation, $\frac{1}{M}\sum_{m=1}^{M} d_{pos,m,c}^2(t) = d_{drift,c}^2(t) + d_{fluc,c}^2(t)$, and convective right-edge acceleration satisfies $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Lieb–Thirring spectral concentration bounds for the fluctuation Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ (since $\mathbf{E}'^\top \mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the Lieb–Thirring spectral concentration ratio [13]:*
$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M} \lambda_m^{\gamma+1/2} + \epsilon_{LT}}, \quad \epsilon_{LT} = 10^{-12}, \tag{4}$$
is radially scale-invariant ($R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$ for $c > 0$ as $\epsilon_{LT} \to 0^+$) and satisfies for any $\mathbf{E}' \neq \mathbf{0}$:
$$\frac{1}{M-1} \leq R_{LT}(\mathbf{E}') \leq 1, \tag{5}$$
attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across $M - 1$ modes, and 1 iff $\operatorname{rank}(\mathbf{E}') = 1$.

Proof. For each coordinate c, expanding $\frac{1}{M}\sum_{m=1}^{M}(\bar{e}_c + e'_{m,c})^2$ with $\sum_{m=1}^{M} e'_{m,c} = 0$ yields the orthogonal energy identity; Cauchy–Schwarz on $\tilde{e}_{M,c} - \tilde{e}_{1,c} = e'_{M,c} - e'_{1,c}$ gives $d_{\text{accel},c}^2 \leq 2(e'^2_{M,c} + e'^2_{1,c}) \leq 2M\, d_{\text{fluc},c}^2$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^\top \mathbf{1}_M) = \mathbf{0}$ forces $\lambda_1 = 0$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \leq \cdots \leq \lambda_M$ with $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^{M} \lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$, which yields (5). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

In Stage 4 (Fig. 1), the 224 fluid-dynamic dials are monitored by an array of independent binary circuit breakers calibrated strictly on nominal data:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1,\dots,224\}$ (and $j \in \{1,\dots,128\}$ in the 128-bit sub-block panel) is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into a non-negative excess score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any drifting dial individually without desensitizing the remaining dials, each tripwire τ_j scales its 98.5th calibration percentile by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\text{cal}}\, \gamma_j^{\text{tr}\to\text{cal}}$:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma_j^{\text{cal}}\, \gamma_j^{\text{tr}\to\text{cal}}, \tag{6}$$
where γ_j^{cal} equals the ratio of second-half to first-half calibration 98th percentiles whenever that ratio exceeds 1.35 (and 1 otherwise), and $\gamma_j^{\text{tr}\to\text{cal}}$ equals $(m_j + s_j)/(1.5\, s_{\text{group},j})$ whenever the dial's calibration scale exceeds $1.5\times$ its perspective-group median $s_{\text{group},j}$ by more than $1.25\times$ (and 1 otherwise).
2) *32 fast right-edge onset relays:* To catch newly entering faults in the terminal sub-block $[48, 64]$ without waiting for earlier prefixes to saturate, each coordinate $c \in \{1,\dots,32\}$ defines a stricter edge switch $b_{p,c}^{\text{edge}}(t) = \mathbb{I}\big(z_{p,c}(t) >$

$q_{0.993}(z_{p,c,\text{cal}})\,\gamma_{p,c}^{1.25}\big)$ across perspectives p and fires its onset relay $r_c(t) \in \{0,1\}$ when the terminal sub-block ($m = 4$) coincides with convective acceleration, turbulent fluctuation, or the third sub-block ($m = 3$):

$$r_c(t) = b^{\text{edge}}_{\text{pos},4,c}(t) \wedge \big(b^{\text{edge}}_{\text{accel},c}(t) \vee b^{\text{edge}}_{\text{fluc},c}(t) \vee b^{\text{edge}}_{\text{pos},3,c}(t)\big). \quad (7)$$

3) *Causal trailing vs. prefix-routed buffered attribution and Boolean quorums:* Under *strictly causal trailing attribution*, window switches at index t are emitted solely at the right endpoint $\tau = t + S - 1$, which observes only $\mathbf{X}[t - C : \tau, :]$ ($\mathcal{F}_\tau$-measurable, with structural emission lag $S - 1 = 63$ steps vs. empirical detection delay $\Delta t = t_{\text{alm}} - t^*$, Fig. 2). Under *retrospective buffered attribution*, each perspective group votes only over the exact time steps inside $[t, t + S)$ that it physically observes: $[t, t + S_m)$ for sub-block $m \in \{1, 2, 3, 4\}$, $[t, t + S)$ for mean drift and fluctuation, and $[t + S_3, t + S_4)$ for right-edge acceleration. Summing active switches at timestamp τ yields counts $N_{224}(\tau) = \sum_{j=1}^{224} \bar{b}_j(\tau)$, $N_{128}(\tau) = \sum_{j=1}^{128} \bar{b}_{128,j}(\tau)$, and $N_{\text{edge}}(\tau) = \sum_{c=1}^{32} \bar{r}_c(\tau)$, with combined count $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ used for threshold-free PR-AUC ranking. Binary alarms trigger when any counter exceeds its nominal calibration quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \quad (8)$$

where $K_{\text{edge}} = \max\big(1.25\, q_{0.999}(N_{\text{edge,cal}}),\, 2.5\big)$, *224-Bit Breaker* sets $K_{224} = \max\big(1.12\, q_{0.995}(N_{224,\text{cal}}),\, 5.0\big)$ and $K_{128} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 3.5\big)$, and *224-Bit Strict* sets $K_{224} = \max\big(1.18\, q_{0.9985}(N_{224,\text{cal}}),\, 5.5\big)$ and $K_{128} = \max\big(1.20\, q_{0.9985}(N_{128,\text{cal}}),\, 4.0\big)$.

1D Scalar Collapse Controls (for Ablation Comparison): To isolate the gains of the 224-bit panel against legacy 1D scalar collapse on the identical trained backbones (Section VI), we evaluate the coordinate saliency gate $g_d(\mathbf{v}) = (1 - \alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ with softmax weights $w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$) . Single-horizon TS-JEPA scores $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$, whereas 1D Soliton-Reynolds JEPA scores the multi-horizon envelope $\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ modulated by $\mathcal{S}_{\text{Sol-Rey}}(t) = \mathcal{S}_{\text{env}}(t)\big(1 + \alpha_{\text{sol}} \max(0, \tanh(0.5\, z_{\text{LT}}(t)))\big)$ ($\alpha_{\text{sol}} = 0.12$, or $\alpha_{\text{sol}} = 0$ for the unmodulated envelope), calibrated via scalar Generalized Pareto Distribution (GPD) tail fitting in SPOT [8], [38], [49] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained on nominal sequences using AdamW [50] (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32), with identical training budgets and zero per-dataset hyperparameter tuning across families.

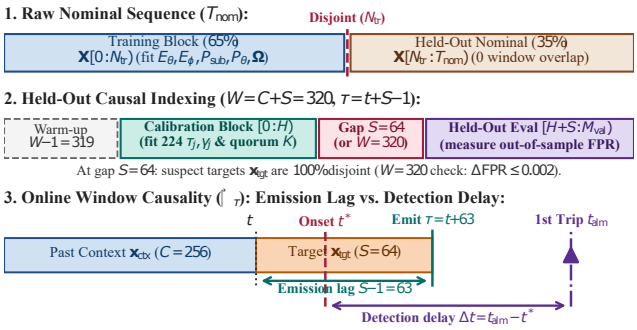


Fig. 2. Held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorum K, and (Row 3) score-emission lag versus fault-detection delay.

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample alarm calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters, neural weights, and shrinkage whitening matrices $\mathbf{\Omega}$ are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W - 1 = 319$ observations provide context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window timestamps. Splitting at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ (used to fit the 224 per-dial tripwires τ_j, hijack guards γ_j, and Boolean quorum K, or scalar τ_{SPOT} for 1D baselines) and the *held-out evaluation block* $[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $[H : H+S)$ so the last calibration target and first evaluation target are 100% disjoint. A full-window isolation check ($G = W = 320$, excluding $[H : H + 320)$ across the seven streams with $M_{\text{val}}/2 > 420$) confirms that shared passive context changes causal held-out FPR by less than 0.25 percentage points.

b) Independent Statistical Units, Multi-Seed Replication, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across multi-seed replications ($\{42, 123, 456, 7, 2024\}$ for 1D baselines and $\{42, 123, 456\}$ for 224-bit circuit-breaker and 141-stream benchmarks). Seeds are averaged within each stream first (SD_{seed}), stream-level paired tests treat the telemetry stream ($n = 33$, $n = 19$, or $n = 141$) as the unit of analysis, and domain-level tests treat the dataset family ($N = 7$ or $N = 12$) as the independent cluster:

1) *Paired Wilcoxon signed-rank test and false-discovery-rate adjustment:* The Wilcoxon signed-rank test evaluates whether the median paired difference across streams is zero without Gaussian assumptions. We adjust p-values via

the Benjamini–Hochberg false-discovery-rate (FDR) procedure [51] ($q_{\text{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; $2{,}000$ resamples), and cross-check at the dataset-cluster level via exact sign-flip permutation tests and zero-tuning LODO sensitivity.

2) *Two One-Sided Tests (TOST) for fixed-backbone regularizer equivalence:* We use TOST [15] to test whether post-hoc scalar soliton modulation alters aggregate Point-F1 relative to the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\text{env}}(t)$ on the identical trained backbone ($n = 19$ streams), establishing equivalence when $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ across $\Delta \in \{\pm 0.015, \pm 0.020, \pm 0.030\}$ (Table II(a)).
3) *Distributional tail and drift diagnostics (γ_2 and D_{KS}):* On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$, distinguishing heavy-tailed *leptokurtic* ($\gamma_2 > 0$) from light-tailed *platykurtic* ($\gamma_2 < 0$) residuals; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0,1]$, quantifying distributional shift between calibration and held-out nominal blocks.

B. Baseline Fairness Checks, Cohort Structure, and Reproducibility

Two fairness audits verify that reconstructive baseline results under the unified $W = 320$ pipeline are not artifacts of window length or implementation: (i) evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their native published window sizes across five streams raises Point-F1 to 0.164 and 0.218, but observation residuals remain severely heavy-tailed ($\gamma_2 > 98$ vs. 3.1 for TS-JEPA) with high nominal FPR (28.4%–44.1%); and (ii) reconstructors replicate their published point-adjusted F1 on SMD and MSL, while Table III(d) provides a secondary 130-stream, 11-dataset cross-check against NCAD, DCdetector, and Anomaly Transformer.

Experiments span five cohorts: *Cohort 1* (33 primary streams: 19 non-GHL and 14 GHL, Table III(a)–(c)); *Cohort 2* (19-stream multi-perspective circuit-breaker ablation, 14-arm matched-backbone factorial comparison, and operating-curve suite, Table II(a),(c) and Figs. 3(a) and 4(b)); *Cohort 3* (6-stream single-channel quiet-fault and hybrid-monitor suite, Fig. 3(b)); *Cohort 4* (5-stream horizon-latency sweep across $S \in \{1, 4, 16, 32, 64\}$ and dyadic front-end ablation, Fig. 4(a) and Table II(b)); and *Cohort 5* (141-stream, 3-seed benchmark across 12 multivariate datasets comparing 1D JEPAs against the 224-bit circuit-breaker panel plus the 130-stream baseline screen, Tables III(d) and V(a)). At $K = 25$ channels, online parameter counts are matched (272k for TS-JEPA, 312k for Soliton-Reynolds JEPA, 304k–307k for TCN observation forecasters, 649k for TimesNet, and 206k for TranAD; the 224-bit circuit-breaker panel adds zero trainable parameters). Runs executed on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB memory); code is at https://github.com/quaid281/NCAD.

C. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [16], [40], we evaluate models across three tiers: (i) *primary accuracy and false-alarm outcomes* comprising realized held-out nominal FPR (FPR_{hold}), unadjusted Point-F1, and threshold-free PR-AUC; (ii) *operating characteristics* evaluating event recall across autonomous circuit-breaker operating points (*224-Bit Strict* and *224-Bit Breaker*), deployable calibration quantiles (q^*, Fig. 4(b)), and post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)); and (iii) *secondary operational endpoints* reporting unweighted Dataset-Macro Point-F1, event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: 1D Scalar Collapse vs. 128-Bit and 224-Bit Multi-Perspective Circuit Breakers

What unlocks the accuracy and speed of Soliton-Reynolds JEPA on the identical trained backbone? Table II(a) and Fig. 3(a) trace the step-by-step progression across all 19 non-GHL streams. First, under 1D scalar scoring, the unmodulated multi-horizon whitened envelope (α_{sol}=0) and the Lieb–Thirring soliton-modulated score (α_{sol}=0.12) achieve strong false-alarm suppression (0.6%–0.7% causal FPR) and satisfy TOST equivalence within $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 2.7\times10^{-5}$), but collapsing 128 whitened coordinates into a single 1D score allows isolated drifting coordinates to inflate the scalar threshold, restricting causal Point-F1 to 0.052–0.055 and delay to 246.6–248.0 steps. Second, replacing 1D collapse with independent per-coordinate binary switches over the 128 sub-block magnitudes (*128-Bit Sub-Block*) immediately lifts causal Point-F1 from 0.052 to 0.129, buffered Point-F1 from 0.137 to 0.192, and causal event recall from 39.8% to 59.0%. Third, expanding the same 32 latent coordinates into $7\times32 = 224$ fluid-dynamic dials (adding Reynolds mean-flow drift, turbulent fluctuation, and convective right-edge acceleration) and 32 fast right-edge onset relays in *224-Bit Breaker* raises causal Point-F1 to 0.138 (17 wins / 0 losses, $p = 1.5\times10^{-5}$), buffered Point-F1 to 0.228 ($p = 0.026$), buffered PR-AUC from 0.370 to 0.411, and event recall to 60.7% causal (63.1% buffered) while cutting detection delay from 246.6 to 105.9 steps causally (402.7 to 71.8 steps buffered). Finally, adding a soft temporal and cross-perspective attention pooling gate over the 224 dials (*224-Bit + Soft Attn*) offers no further gain (0.137 causal and 0.226 buffered F1), confirming that discrete Boolean switches prevent cross-coordinate dilution without requiring continuous attention weights.

B. Ablation 2: Single-Channel Quiet Faults, Input Gating, and Hybrid Monitoring

When one sensor out of K freezes at a nominal value while the remaining $K-1$ channels fluctuate normally, the frozen channel perturbs the pooled pre-LayerNorm vector by

TABLE II
MULTI-PERSPECTIVE CIRCUIT-BREAKER ABLATION AND MECHANISM-DISCRIMINATION STUDIES (n=19 STREAMS; EVENT RECALL (ER)).

(a) 1D Collapse vs. 224-Bit Circuit-Breaker Progression (n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Scoring / Panel Variant	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
Sol.-Rey. 224-Bit Breaker	**.138***	**.228**†	**.312**	**.411**	**.607**	**.631**	**105.9**	**71.8**
Sol.-Rey. 224-Bit Strict	.116*	.190	**.312**	**.411**	.534	.545	248.0	97.3
Sol.-Rey. 224-Bit + Soft Attn	.137	.226	.311	.410	**.607**	.628	106.8	73.4
Sol.-Rey. 128-Bit Sub-Block	.129	.192	.310	.402	.590	.525	157.4	273.9
Sol.-Rey. 1D (α_{sol}=0.12)	.052	.137	.291	.370	.398	.377	246.6	402.7
Sol.-Rey. 1D (α_{sol}=0 Env.)	.055	.138	.291	.368	.398	.384	248.0	398.2

* Wilcoxon vs. 1D Sol.-Rey.: 224-Bit Breaker 17/0 wins ($p=1.5\times10^{-5}$), 224-Bit Strict 16/1 wins ($p=3.1\times10^{-5}$); † buffered F1 $p=0.026$. Fixed-backbone 1D $\alpha_{\text{sol}}=0.12$ vs. $\alpha_{\text{sol}}=0$ satisfies TOST equivalence at $\Delta=\pm0.020$ ($p_{\text{TOST}}=2.7\times10^{-5}$).

(b) Dyadic Front-End (n=5)

Stream (F1 / PR)	Std. Conv	Dyadic	ΔF1 / ΔPR
SMAP P-3	**0.301** / 0.280	0.115 / **0.296**	−0.186 / +0.017
MSL M-1	0.014 / **0.491**	**0.016** / 0.449	+0.002 / −0.043
SMD machine-1-2	0.000 / 0.028	**0.068** / **0.221**	+0.068 / +0.193
Daphnet S01R01E1	**0.179** / **0.103**	0.130 / 0.066	−0.048 / −0.037
GECCO water-quality	**0.097** / 0.161	0.062 / **0.282**	−0.035 / +0.121
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	−0.040 / +0.050

(c) Matched-Backbone Architectural Comparison (n=19 **Streams**)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	Auto.	q^*=0.05	f^*=0.01	f^*=0.05
Soliton-Reynolds 224-Bit Breaker	2.30	0.179	0.017	0.132	0.138	0.228	**0.312**	**0.411**	**0.607**	0.774	**0.584**	**0.728**
Soliton-Reynolds 224-Bit Strict	2.30	0.179	0.011	0.098	0.116	0.190	**0.312**	**0.411**	0.534	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Soliton-Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Soliton-Reynolds (α_{sol}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Hard-Target Copy	1.16	**0.167**	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

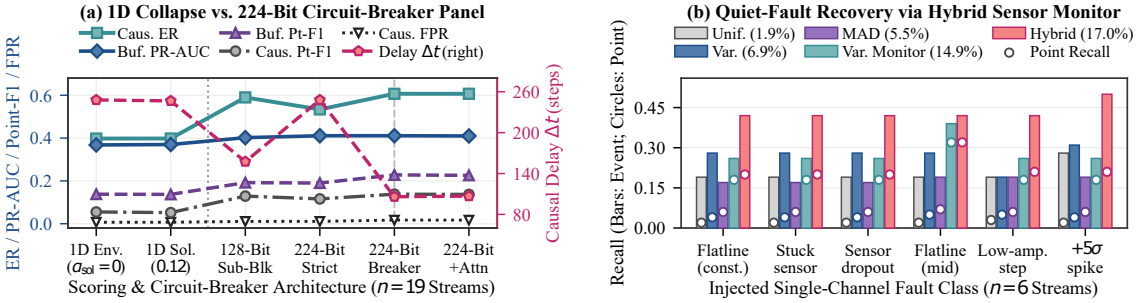


Fig. 3. Multi-perspective circuit-breaker progression and quiet-fault recovery. **(a)** Causal ER, buffered PR-AUC, buffered Point-F1, causal Point-F1, causal held-out FPR (left axis), and causal detection delay Δt (right axis) across 1D scalar collapse ($\alpha_{\text{sol}} \in \{0, 0.12\}$), the 128-bit sub-block panel, 224-Bit Strict, 224-Bit Breaker, and 224-Bit + Soft Attention (n=19 streams). **(b)** Injected single-channel ER (bars) and point recall (circles) across uniform TS-JEPA (Unif.), context-variance (Var.) and MAD input gates, the standalone rolling variance-collapse monitor (Var. Monitor), and the hybrid union detector (Hybrid; nominal FPR in parentheses, n=6 streams, three seeds).

only $O(1/K)$, and LayerNorm rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$ using the remaining active channels. Pre-weighting input channels by context variance lowers macro Point-F1 across four multivariate streams from 0.118 to 0.077, so the deployed pipeline retains uniform input encoding. Across six single-channel fault classes injected into six multivariate streams (Fig. 3(b)), uniform TS-JEPA detects 27.8% of abrupt $+5\sigma$ spikes, but on persistent quiet faults (flatlines, stuck sensors, and zero dropouts) event recall falls to 19.4% and point recall drops below 3%. Simple observation-space monitors catch these complementary failures: a maximum channel deviation monitor and a rolling variance-collapse monitor (tracking how far each channel's 32-step trailing standard deviation drops below its nominal median) each recover 25%–39% of quiet-fault events at 14%–15% FPR. Taking the logical union of TS-JEPA alarms and the rolling variance-collapse monitor lifts event recall to 41.7%–50.0% across all six fault classes at 17.0% nominal FPR.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Table II(b) trains TS-JEPA with the identical 3-layer causal TCN encoder across five streams, comparing the 4-shell Fourier filterbank against a standard linear convolutional stem. Enabling the dyadic front-end raises macro PR-AUC from 0.213 to 0.263 (led by SMD and GECCO) and lowers validation residual kurtosis from 1.41 to 1.07, while macro Point-F1 under 1D thresholding decreases from 0.118 to 0.078 on SMAP P-3. Frequency-shell splitting thus improves score ranking and tail conditioning.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 14 model arms across all 19 non-GHL streams under identical splits, training budgets, and matched parameter counts in Table II(c) isolates five architectural factors:

1) *Forecasting vs. reconstructing raw observations (TCN Obs. Forecaster vs. TCN Obs. Autoencoder):* Pairing the same

causal dilated TCN encoder with a multilayer perceptron decoder to either forecast the 64-step target window or reconstruct the 320-step window in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), calibration drift ($D_{\text{KS}} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical.

2) *LayerNorm bottleneck in observation space (TCN Obs. Forecaster (LN)):* Adding a $D = 32$ LayerNorm bottleneck and target-window LayerNorm to the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal FPR to 11.7%, but targeting raw sensor waveforms leaves calibration drift high ($D_{\text{KS}} = 0.376$) and drops causal Point-F1 to 0.084.
3) *Learned latent target projection, multi-horizon whitening, and 224-bit circuit breakers (TS-JEPA and Soliton-Reynolds 224-Bit):* Moving the prediction target into the learned $D = 32$ latent space cuts calibration drift nearly in half ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179 in Soliton-Reynolds JEPA, paired Wilcoxon $p \leq 0.001$), while the 224-bit circuit-breaker panel lifts buffered PR-AUC to 0.411, buffered Point-F1 to 0.228, and causal event recall to 0.607 at 1.7% held-out FPR.
4) *Nonlinear TCN encoding vs. linear PCA and tree isolation (PCA Linear Forecaster (LN) and Isolation Forest):* Replacing the nonlinear TCN with 32 principal components and ridge regression bounds causal FPR to 2.1% through LayerNorm, but causal Point-F1 collapses to 0.017 (and Isolation Forest reaches only 0.149 causal event recall), confirming the need for learned nonlinear temporal features.
5) *Future prediction vs. context autoencoding, EMA stability, and ℓ_2 scoring (JEPA Context Autoencoder, JEPA Hard-Target Copy, and TS-JEPA (ℓ_2 Scoring)):* Context autoencoding in latent space keeps causal FPR low (1.7%) but drops causal Point-F1 from 0.161 to 0.112 and PR-AUC from 0.278 to 0.235. Direct weight copying ($\zeta = 0$) or Euclidean ℓ_2 scoring still holds causal FPR below 2% (0.9% and 1.8%), confirming that false-alarm reduction stems from the normalized latent representation.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 primary continuous telemetry streams and the 141-stream archive-wide benchmark under the chronological held-out protocol: Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, Panel (c) per-dataset breakdowns including anomaly event counts (E) and both causal and buffered held-out nominal FPRs, and Panel (d) the 130-stream and 141-stream archive evaluations. Across both attribution regimes, pairing multi-horizon Soliton-Reynolds JEPA with the 224-bit multi-perspective circuit-breaker panel achieves the highest buffered Point-F1 (0.228), buffered PR-AUC (0.411), causal PR-AUC (0.312), causal event recall (0.607), and Dataset-Macro Point-F1 (0.122), while keeping held-out false alarms an order of magnitude below observation-space reconstructors (1.1%–1.7% causal and 9.8%–13.2% buffered vs. 22.2%–30.7% for TimesNet and TranAD).

Suppressing False-Alarm Flooding Relative to Reconstructors: On the 19 non-GHL streams (Table III(a),(b)), latent prediction cuts held-out nominal FPR three-fold under buffered attribution (7.8%–9.8% for Soliton-Reynolds 1D and 224-Bit Strict vs. 30.0%–30.7% for TimesNet and TranAD, $p \leq 0.045$) and by more than an order of magnitude under strictly causal streaming (0.6%–1.7% for Soliton-Reynolds variants and 2.4% for TS-JEPA vs. 22.2%–24.3% for reconstructors, $p = 0.010$). This separation holds across all $N = 7$ dataset families ($p = 0.016$) and under zero-tuning LODO evaluation (0.3%–2.3% causal FPR for JEPAs vs. 12.1%–18.4% for reconstructors). In Table III(c), Soliton-Reynolds 224-Bit achieves 0.7% causal and 4.7% buffered FPR on Daphnet (0.223 buffered F1 vs. 0.145 for TS-JEPA), 0.199 buffered F1 on MSL (surpassing TS-JEPA at 0.161 and TimesNet at 0.158), 0.354 buffered and 0.351 causal F1 on SMAP at 38.2-step causal delay, and 0.063 buffered F1 on Genesis.

Overcoming 1D Score Collapse Across 19 Primary and 141 Archive Streams: Under legacy 1D scalar scoring, Soliton-Reynolds JEPA trades lower Point-F1 (0.050 causal, 0.148 buffered) for ultra-conservative firing because one or two out-of-sample drifting coordinates (*hijacker dials*) inflate the scalar threshold. Replacing 1D scalar collapse with the 224-bit multi-perspective circuit-breaker panel resolves this trade-off on the exact same trained networks (Tables II(a), III(a),(b),(d), and V(a)). On the 19 primary non-GHL streams, *Soliton-Reynolds 224-Bit Strict* and *224-Bit Breaker* lift buffered PR-AUC from 0.331–0.370 to 0.411, raise buffered Point-F1 to 0.190–0.228 ($p = 0.026$), more than double causal Point-F1 from 0.052 to 0.116–0.138 (17 wins / 0 losses, $p = 1.5\times10^{-5}$), lift causal event recall to 53.4%–60.7% (63.1% buffered), and cut detection delay from 246.6 to 105.9 steps causally (402.7 to 71.8 steps buffered). Across all 141 qualifying multivariate streams in 12 datasets at three seeds (Tables III(d) and V(a)), 3-horizon Speculative-JEPA and Speculative-Alien-JEPA first improve 1D PR-AUC over single-horizon TS-JEPA (0.184 vs. 0.165 causally; 0.253–0.256 vs. 0.213 buffered, $p \leq 0.0042$). Pairing the multi-horizon predictor with the 224-bit circuit-breaker panel (*Speculative-224Bit-Strict* and *Speculative-224Bit-Breaker*) then lifts causal PR-AUC to 0.197 ($p = 0.014$ vs. TS-JEPA), nearly doubles causal Point-F1 from 0.067–0.077 to 0.125–0.139 (97 wins / 36 losses over TS-JEPA, $p = 8.3\times10^{-11}$; 114 wins / 18 losses over 1D Speculative-Alien-JEPA, $p = 1.2\times10^{-16}$), raises causal event recall from 48.0%–50.7% to 64.8%–72.7%, and halves causal detection delay from 115.4–125.4 to 55.3–78.0 steps. Under buffered attribution across the 141 streams, the 224-bit panel lifts PR-AUC to 0.261 ($p = 0.0031$), raises Point-F1 to 0.177–0.180 ($p \leq 2.6\times10^{-4}$), raises event recall to 66.0%–68.5%, and lowers held-out FPR below 1D Speculative-Alien-JEPA (16.5% vs. 18.0%) at 46.6–54.0-step delay. On GHL ($\pi = 0.013$, Table III(c)), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1), showing that low FPR without recall does not indicate detection.

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

TABLE III
HELD-OUT MULTI-STREAM TELEMETRY BENCHMARK ACROSS ATTRIBUTION REGIMES AND DATASETS (BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-perspective fluid-dynamic circuit-breaker architectures												
Soliton-Reynolds 224-Bit Breaker	0.132	**0.228**	**0.411**	**0.631**	71.8	0.017	0.138	**0.607**	105.9	**0.002**	0.000	**0.122**
Soliton-Reynolds 224-Bit Strict	0.098	0.190	**0.411**	0.545	97.3	0.011	0.116	0.534	248.0	**0.002**	0.000	0.098
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Soliton-Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Soliton-Reynolds (α_{sol}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	F1_{all}	**224-Bit**	**TS-JEPA**	**Sol. 1D**	**TimesNet**	**TranAD**	**224-Bit**	**TS-JEPA**	**Sol. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.223	0.145	0.143	**0.229**	0.144	**.007/.047**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.001/.002**	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.199**	0.161	0.081	0.158	0.151	.014/.131	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.017	**0.119**	0.008	0.000	0.000	**.000/.000**	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: 95% stream-bootstrap CIs (n=19) for (a) Buffered FPR/F1: 224-Bit Breaker [.068, .198]/[.142, .321], 224-Bit Strict [.044, .156]/[.112, .276], TS-JEPA [.048, .145]/[.113, .319], Sol.-Rey. 1D [.032, .137]/[.081, .229], TimesNet [.162, .446]/[.098, .315], TranAD [.177, .445]/[.053, .232]; (b) Causal FPR/F1: 224-Bit Breaker [.008, .027]/[.068, .224], 224-Bit Strict [.005, .018]/[.051, .195], TS-JEPA [.007, .046]/[.063, .287], Sol.-Rey. 1D [.003, .011]/[.013, .097], TimesNet [.123, .375]/[.109, .326], TranAD [.119, .337]/[.058, .241]. Rate-matched random alerter $\text{ER}_{\text{rand}}(\text{FPR}_{\text{hold}})$: Sol.-Rey. 224-Bit .281 (vs. .607 actual), Sol.-Rey. 1D .272 (vs. .478 actual), TS-JEPA .283 (vs. .487 actual), TimesNet .600 (vs. .588 actual). **(d) Archive-Wide Benchmarks:** *130-Stream Screen (Pt-F1, macro/stream):* NCAD .058/.088; DCdetector .070/.097; Anomaly Trans. .091/.082; TimesNet .129/.151; TranAD .156/.152; TS-JEPA **.160/.186**. *141-Stream, 3-Seed Benchmark (12 Datasets; Causal/Buffered Pt-F1, PR-AUC [ER, Δt]):* TS-JEPA .067/.142, .165/.213 [.480/.459, 115.4/117.9]; Speculative-JEPA .071/.155, .183/.256 [.497/.565, 123.0/65.8]; Speculative-Alien-JEPA .077/.153, .184/.253 [.507/.567, 125.4/65.1]; **Speculative-224Bit-Strict** .125/.177, **.197**/**.261** [.648/.660, 78.0/54.0]; **Speculative-224Bit-Breaker** **.139**/**.180**, **.197**/**.261** [**.727**/**.685**, **55.3**/**46.6**] (Table V(a)).

1) *Favorable operating regime:* Multi-channel spacecraft, server, and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet, SMD, Exathlon) with localized transients, where the 224-bit circuit-breaker panel maximizes Point-F1 and event recall at low false-alarm rates.
2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response (such as hourly Metro where 63 steps = 63 h), requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring the hybrid union detector (combining TS-JEPA with a rolling variance-collapse monitor) when single-channel coverage is mandatory (Section VI-B).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling calibration rules.

B. Component-Level Analysis of Soliton-Reynolds and the 224-Bit Breaker

Tables II(a),(c) and III(a),(b) isolate how each architectural stage contributes to overall performance. First, *multi-depth TCN tapping and barycenter centering* remove common-mode baseline shift, lowering calibration drift (D_{KS} from 0.214 to 0.179) and cutting the inter-block drift share of excess causal FPR four-fold (71.3% to 18.7%, Table V(b)). Second, *nested multi-horizon rollout and calibration-stabilized Mahalanobis whitening* capture sub-horizon transients across $S_m \in \{16, 32, 48, 64\}$, lifting causal event recall at a 1% FPR budget from 0.395 to 0.584. Third, while *post-hoc 1D Lieb–Thirring soliton modulation* ($\alpha_{\text{sol}} = 0.12$) leaves 1D Point-F1 and PR-AUC statistically unchanged under TOST ($p_{\text{TOST}} < 10^{-4}$), *expanding the 32 whitened coordinates into the 224-bit fluid-dynamic circuit-breaker panel with 32 fast right-edge onset relays* eliminates hijacker-dial threshold inflation, doubling causal Point-F1 and halving detection delay.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III shows why the 224-bit circuit-breaker panel is essential for real-time streaming. Without 64-step window averaging, causal 128-dimensional whitened residuals have a heavier right tail ($\gamma_2 = 2.30$) than buffered scores ($\gamma_2 = 1.25$). Under legacy 1D scalar scoring, hijacker dials push the scalar threshold high onto that tail so that 1D Soliton-Reynolds JEPA alerts on only $\hat{p}_{\text{alert}} = 2.4\%$ of test steps (4.0% true-positive rate (TPR) at 0.6% held-out FPR), flagging only isolated peak steps inside each event (0.050–0.052 causal F1). By monitoring each fluid-dynamic dial with its own hijack-guarded switch $b_j(t)$ and triggering immediately when a terminal sub-block and right-edge convective acceleration relay $r_c(t)$ coincide in (7), *224-Bit Breaker* stays active across the true fault duration (0.138 causal F1, 60.7% event recall, and 105.9-step delay at 1.67% held-out FPR). Against an analytical rate-matched i.i.d. Bernoulli random alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$ (Table III Note), random firing at 224-Bit Breaker's held-out nominal rate catches only 28.1% of events versus 60.7% actual, whereas for TimesNet (24.3% FPR) random firing alone reaches 60.0%, exceeding TimesNet's actual 58.8%.

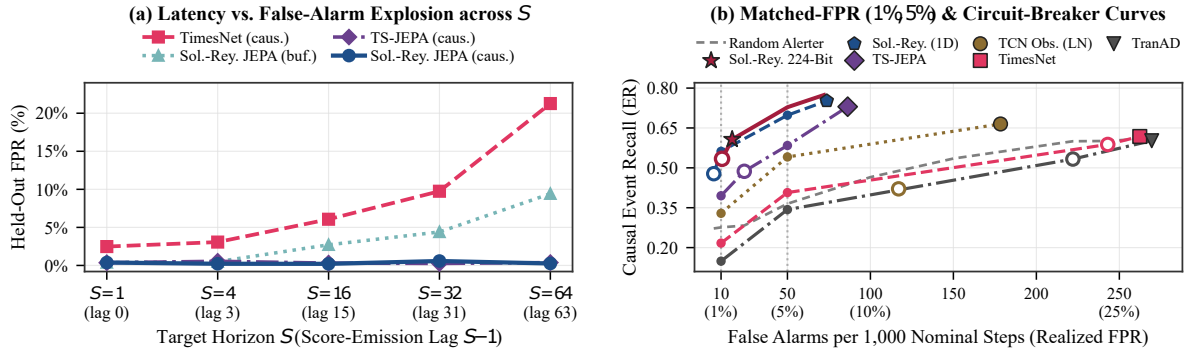


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched score-emission latency $S-1 \in \{0, 3, 15, 31, 63\}$ steps (five streams, three seeds). **(b)** Causal event recall versus false alarms across 19 non-GHL streams at equalized test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides), autonomous 224-Bit Strict (open crimson circle) and 224-Bit Breaker (filled crimson star) operating points, and deployable calibration quantiles up to q^*=0.05, plotted against a rate-matched Bernoulli random alerter.

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR** f^*=.01	f^*=.05		f^*=.01	f^*=.05
Sol.-Rey. 224-Bit	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Soliton-Rey. (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Sol.-Rey. (α_{sol}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Sol.-Rey. 224-Bit	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Soliton-Rey. (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Sol.-Rey. (α_{sol}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 primary SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL streams into in-range contextual ($\max|z| \leq 5\sigma$) vs. out-of-range point ($\max|z| > 5\sigma$) causal events.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

Target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger (Section IV-D). To evaluate detection quality versus latency against baselines at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero lag) and $S = 4$ (3-step lag), TS-JEPA keeps causal FPR at 0.4%–0.6% (vs. 2.5%–3.1% for TimesNet) while maintaining higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162) and short detection delays (Δt = 57.8–85.2 steps). As S increases from 16 to 64, TimesNet causal FPR climbs from 6.1% to 21.3%, whereas latent predictors remain bounded ($\leq$ 2.0%). Consequently, $S = 64$ suits high-rate sensors ($\geq$ 1 Hz, such as 64 Hz Daphnet where 63 steps = 0.98 s) or slow subsystem trend monitoring (0.1 Hz SMAP, MSL, and GHL, 10.5 min). Low-rate infrastructure streams ($\leq 1\,\text{min}^{-1}$, such as hourly Metro where 63 h lag is prohibitive) and rapid tripwires instead require $S \in \{1, 4\}$.

E. Operating Budgets and Contextual vs. Point Anomalies

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.6%–2.4% for latent predictors, we evaluate three operating-point comparisons:

1) *Autonomous circuit-breaker points and nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):* Without test labels, Soliton-Reynolds 224-Bit Strict (53.4% ER at 1.1% FPR) and 224-Bit Breaker (60.7% ER at 1.7% FPR) dominate the low-FPR frontier; at calibration quantile $q^* = 0.050$, Soliton-Reynolds 224-Bit reaches 77.4% causal event recall (vs. 61.8% for TimesNet and 60.2% for TranAD).
2) *Post-hoc matched realized test-FPR budgets (f^*, Fig. 4(b), Table II(c)):* Equalizing realized test nominal FPR across models (vertical dotted guides in Fig. 4(b)) proves that low false-alarm rates are not an artifact of low alert frequency. At a 1% test FPR budget ($f^* = 0.01$), Soliton-Reynolds 224-Bit detects 58.4% of anomaly events (56.2%–57.8% for 1D Soliton-Reynolds and 39.5% for TS-JEPA),nearly tripling TimesNet (21.7%) and quadrupling TranAD (14.9%). A similar gap holds at $f^* = 0.05$ (72.8% vs. 34.3%–40.7%).
3) *Contextual vs. point anomaly stratification (Table IV):* Out-of-range point anomalies (abrupt spikes exceeding $\pm 5\sigma$) can be flagged even by a memoryless marginal z-score (`ZScoreMax`), whereas contextual anomalies stay within normal sensor bounds ($\max|z| \leq 5\sigma$) and only violate multi-step trajectory structure. In Table IV(a), stratifying the 10 primary SMAP/MSL streams by NASA's official ground-truth taxonomy [4] shows that observation-space detectors collapse on contextual anomalies (0.161–0.317 buffered PR-AUC and 30%–56% recall at $f^* \in \{0.01, 0.05\}$). Soliton-Reynolds 224-Bit leads every contextual metric (0.658 buffered PR-AUC, 75.0% autonomous recall, and 82.0%–92.0% recall at $f^* \in \{0.01, 0.05\}$) while reaching 94.4% recall at $f^* = 0.05$ on NASA point anomalies. Across all 19 non-GHL streams (Table IV(b)), Soliton-Reynolds 224-Bit more than doubles the recall of reconstructors on in-range contextual faults (61.8% vs. 30.0% at $f^* = 0.05$) while also leading on point faults (79.2% vs. 56.3%–61.1%).

F. Why 1D Scalar Calibration Fails vs. 224-Bit Stabilization and Drift Decomposition

Table V synthesizes why 224-bit multi-perspective circuit breakers outperform 1D scalar calibration across all 141

TABLE V
141-STREAM 224-BIT CIRCUIT-BREAKER BENCHMARK AND REGIME-DRIFT DECOMPOSITION.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
(a) 141-Stream Cohort	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.067	.142	.165	.213	.480	.459	115.4	117.9
Speculative-JEPA (1D)	.071	.155	.183	.256	.497	.565	123.0	64.8
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	125.4	65.1
Spec.-224Bit-Strict	.125*	.177‡	**.197**	**.261**	.648	.660	78.0	54.0
Spec.-224Bit-Breaker	**.139**†	**.180**‡	**.197**	**.261**	**.727**	**.685**	**55.3**	**46.6**

	Refitted Causal FPR (n=19)			Excess Share (%)		
(b) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	**Inter-Blk**	**Serial**	**i.i.d.**
TS-JEPA	0.0229	0.0073	0.0033	71.3	18.1	10.5
Soliton-Rey.	**0.0056**	**0.0047**	**0.0020**	**18.7**	59.5	21.8
TimesNet	0.2302	0.0427	0.0035	81.8	17.1	1.1
TranAD	0.2103	0.0464	0.0034	78.3	20.5	1.1

In (a), n=141 streams across 12 datasets (3 seeds): *Wilcoxon vs. TS-JEPA $90/43$ wins (p=3.6×10^{-7}) and vs. 1D Spec.-Alien $98/29$ wins (p=1.8×10^{-11}); †$97/36$ wins vs. TS-JEPA (p=8.3×10^{-11}) and $114/18$ wins vs. 1D Spec.-Alien (p=1.2×10^{-16}); ‡buffered F1 $p\leq2.6\times10^{-4}$ (Strict lowers buffered held-out FPR from 18.0% to 16.5%). In (b), 95% CIs for inter-block drift $FPR_{chr} - FPR_{blk}$: TS-JEPA $+.0156$ $[+.001, +.034]$, Sol.-Rey. $+.0009$ $[-.002, +.004]$, TimesNet $+.1874$ $[+.078, +.305]$, TranAD $+.1639$ $[+.072, +.267]$.

streams (Panel (a)) and decomposes the underlying regime-drift mechanisms across the 19 primary streams (Panel (b)):

1) *Three-stage permutation drift decomposition (Table V(b)):* To separate inter-block regime drift from serial correlation, we pool nominal calibration and evaluation windows across all 19 non-GHL streams and refit thresholds under three splits: chronological (FPR_{chr}), $L = 64$ block permutation (FPR_{blk}, removing macro-block drift while preserving intra-block serial correlation), and pointwise i.i.d. permutation (FPR_{pt}). Under i.i.d. shuffling, every model achieves low false-alarm rates ($FPR_{pt} \leq 0.35\%$). In chronological deployment, inter-block drift ($FPR_{chr} - FPR_{blk}$) accounts for 71.3% of excess causal FPR in single-horizon TS-JEPA and 78.3%–81.8% in reconstructors, whereas in barycenter-centered Soliton-Reynolds JEPA the inter-block drift share drops four-fold to 18.7% ($FPR_{chr} = 0.56\%$ vs. $FPR_{blk} = 0.47\%$).
2) *Per-dial hijack isolation and prefix-horizon voting:* Even when overall barycenter drift is small, one or two individual coordinates can shift between training and calibration. In 1D scalar models, those coordinates hijack the scalar tail; in the 224-bit circuit-breaker panel, calibration-stabilized whitening $\lambda_{m,c}$ and per-dial hijack guards $\gamma_j^{\text{cal}}\gamma_j^{\text{tr}\to\text{cal}}$ isolate drifting dials individually, while prefix-horizon voting in buffered mode prevents 64-step window smearing from dilating isolated short-horizon trips (lowering buffered held-out FPR from 18.0% in 1D Speculative-Alien-JEPA to 16.5% in Speculative-224Bit-Strict while lifting Point-F1 from 0.153 to 0.177, Table V(a)).

VIII. CONCLUSION AND FUTURE WORK

We investigated how Joint-Embedding Predictive Architectures and fluid-dynamic multi-perspective circuit-breaker panels resolve both observation-space false-alarm flooding and 1D scalar score collapse across 33 primary and 141 archive-wide continuous telemetry streams:

1) *224-Bit Fluid-Dynamic Circuit Breakers Eliminate 1D Score Collapse:* Expanding $D = 32$ multi-horizon whitened latent coordinates into $7 \times 32 = 224$ fluid-dynamic dials (128 sub-horizon flow positions, 32 Reynolds mean-flow transport dials, 32 turbulent-fluctuation dials, and 32 convective right-edge acceleration dials) with nominal-only hijack guards and 32 fast right-edge onset relays prevents isolated drifting coordinates from inflating scalar thresholds. This doubles strictly causal Point-F1 ($0.052 \to 0.138$ on n=19 primary streams, $17/0$ wins, $p = 1.5\times10^{-5}$; 0.067–0.077 $\to 0.139$ across all 141 streams, $114/18$ wins, $p = 1.2\times10^{-16}$), lifts causal event recall to 60.7%–72.7%, and cuts detection delay in half ($246.6 \to 105.9$ steps on n=19 and $125.4 \to 55.3$ steps on n=141).
2) *Factorial Mechanism Isolation and Multi-Horizon Whitening:* Matched-encoder ablations separate score bounding from representation learning: LayerNorm bottlenecks compress residual kurtosis (γ_2 from 6.10 to 0.91), learned latent projections cut calibration drift in half (D_{KS} from 0.376 to 0.214 in TS-JEPA), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to 0.179 while cutting the inter-block drift share four-fold (71.3% to 18.7%). On the fixed backbone, post-hoc 1D Lieb–Thirring soliton modulation satisfies TOST equivalence ($p_{\text{TOST}} < 10^{-4}$), and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag.
3) *Contextual-Anomaly Dominance and Hybrid Quiet-Fault Coverage:* Soliton-Reynolds 224-Bit nearly triples reconstructor event recall at a 1% test FPR budget (58.4% vs. 21.7% for TimesNet) and leads on NASA contextual anomalies (0.658 buffered PR-AUC and 82.0%–92.0% matched-FPR recall), while a hybrid sensor-variance monitor recovers single-channel flatlines and dropouts (41.7%–50.0% event recall), motivating online conformal circuit-breaker adaptation for non-stationary process ramps.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.
[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.
[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.
[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.
[5] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.
[6] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.
[7] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.
[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.
[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.
[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.
[11] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.
[12] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.
[13] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[14] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[15] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.

[16] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[17] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[18] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[19] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[20] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[21] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[22] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[23] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[24] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[25] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[26] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[27] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[28] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[29] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[30] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[31] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[32] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[33] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[34] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[35] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[36] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[37] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[38] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[39] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[40] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[41] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[42] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[43] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[44] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[45] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[46] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[47] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[48] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[49] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[51] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.